

SHRODDHA CHAKRABARTY

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OBJECTIVE:

Driven and passionate Bioengineering undergraduate with a strong interest in epidemiology, neuroscience, biomedical engineering, and public health. Particularly fascinated by the use of data-driven insights to combat neurological disorders, the development of biomedical solutions for disease prevention, and the implementation of public health strategies to enhance healthcare accessibility. Seeking research-oriented roles or internships where I can leverage my technical skills, research experience, and leadership capabilities to contribute meaningfully to impactful healthcare innovations.

CORE SKILLS & TECHNICAL PROFICIENCIES:

- Research & Laboratory Skills: Blood testing and component separation, Spectroscopy (UV-Vis, Fluorescence), Electron Microscopy, X-Ray Diffraction, Gait Analysis, Pedobarograph-based studies, drop casting method, electrospinning.
- Programming & Analytics: Python (NPTEL Joy of Python Course Certified), MATLAB, AI and ML foundations, Data Analysis and Visualization.
- Project Management: Finance and Documentation, PR and Outreach, Technical Documentation.
- Soft Skills: Exceptional communication (Certified by the University of Amsterdam), Content writing, Analytical thinking, Creative problem-solving, and Collaboration.

COMPLETED PROJECTS:

Research & Projects:

1. AI-Powered Mental Health Chatbot

- Developed a fundamental-level AI chatbot to bridge mental health gaps by providing users with accessible mental health insights.
- Focused on enhancing user interaction and emotional validation.

2. Contamination of the Environment and Depletion of the Biosphere due to Biomagnification of Harmful Chemicals

- Researched the impact of harmful chemicals on the environment and developed preventive measures.

3. AI-Powered Legal Assistance Platform

- Contributed to the development of a system that offers AI-driven legal support and project automation.

4. Immunotherapy and Precision Medicine for Cancer Treatment: A Comprehensive Review

- Reviewed advancements in immunotherapy and their implications for personalized medicine in cancer treatments.

5. The Functional Role of Microbial Diversity in Organic Matter Turnover in Soil

- Conducted in-depth research on soil microbiology, emphasizing the ecological significance of microbial diversity.

6. Bacterial Endotoxins as Potential Cancer Therapeutics: A Comprehensive Review

- A comprehensive review on Bacterial Endotoxins as Potential Cancer Therapeutics would explore the evolving field of endotoxin-based therapies and their potential to treat cancer. Endotoxins are components of the outer membrane of Gram-negative bacteria, and they have been studied for their immunomodulatory properties.

INTERNSHIP:

INDIAN INSTITUTE OF TECHNOLOGY, INDORE as a summer research intern. I am currently working on PVDF-BTO thin film to measure the dielectric change at different annealed temperatures and generate piezoelectricity for its biomedical application(biosensors).

DELOITTE INTER. In the Transformation and Technology(T&T) sector of **SUPPLY CHAIN** and **NETWORK OPERATIONS**

SRAMAJIBI HOSPITAL as **BLOOD BANK INTERN.** Conducted various **blood tests** and **component separation** procedures, including crossmatching, blood grouping, and centrifugation for component extraction, while ensuring adherence to clinical safety protocols.

- Ensured adherence to safety protocols while maintaining data accuracy.
- Gained hands-on experience in the fundamentals of blood banking and diagnostics.

SANGENE BIOTECH:

To demonstrate practical skills in extracting and analyzing DNA. To understand its applications in biotechnology. To evaluate its role in genetic research and diagnostics. The highlight of the session was a hands-on workshop on extracting and isolation of DNA and doing Polymerase Chain Reaction (PCR). We learned about the principles of DNA structure, safety and ethical considerations. We also learned about the calculations of solutions.

HACKATHONS:

Johns Hopkins University hackathon. Here we worked on Developing a solution that supports personalized healthcare by leveraging data to create tailored treatment plans, predictive models, or patient support tools.

Smart India Hackathon (SIH) PS Code:1285. we made a legal assistant chatbot. It has three stages; the first one the legal document simplifying module, the second module is the drafting legal documentation module and lastly the third module is the consulting module.

WORKSHOPS and Industrial Trainings:

National Institute of Wind Energy (NIWE): To gain firsthand knowledge of the principles and applications of wind energy technology. To explore the research and development activities undertaken by NIWE in the field of wind energy. To learn about the latest advancements and future trends in the wind energy industry. To observe the operation of wind energy equipment and facilities. To foster a deeper appreciation for the potential of wind power as a sustainable energy source.

NeoSky, Bangalore:

To gain a comprehensive understanding of drone technology and its applications. To learn about the manufacturing process of different types of drones. To gain hands-on experience flying drones in a controlled environment.

TVS (Training and Services), Chennai:

The objective of Day 4's visit to TVS in Chennai was to explore the integration of advanced technology in the automotive sector, with a focus on Industry 4.0. we aimed to gain insights into how automation, IoT, AI, and electric motor technology are revolutionizing automotive manufacturing and transforming production lines for greater efficiency.

BSNL RGMTTC, Chennai:

the focus was on understanding the principles of optical fiber technology, its applications, and the methods used to establish network connections. , we aimed to explore how high-speed data transmission is achieved in telecommunications and the operational mechanisms of systems like WLL-M.

VIT Chennai – Hands-on Training on Microcontroller Programming (RP2040):

the primary objective was to dive deep into microcontroller programming, specifically focusing on the RP2040 microcontroller from the Raspberry Pi Foundation. This session aimed to provide practical exposure to embedded systems and the integration of hardware and software. we were tasked with not only understanding the architecture of the RP2040 but also learning how to use its dual-core capabilities to implement multitasking and parallel processing in real- world applications. The hands-on session was designed to help us explore the potential of microcontrollers in embedded systems, learning how software drives hardware functionality and controls physical components like LEDs.

EDUCATION:

Bachelor of Technology (B.Tech) in Bioengineering

Vellore Institute of Technology (VIT), Bhopal Campus

Expected Graduation: 2026

- Relevant Coursework: Biomechanics (including soft tissue mechanics and bone fracture analysis), Finite Element Analysis (with a focus on domain discretization and nodal solutions), Epidemiology (emphasizing disease pattern analysis and preventive measures), Neuroscience (neuroplasticity and brain-computer interfaces), Biomedical Instrumentation (signal processing and imaging techniques), UV-Vis Spectroscopy (quantitative and qualitative analysis methods), and Public Health (policy-making and data-driven health interventions).

EXTRACURRICULAR ACTIVITIES & LEADERSHIP ROLES:

Technical Clubs

- Android Club: Finance and Documentation Lead
- The Null Chapter: PR and Outreach Lead
- Bioengineering Club: Core Member in Project Department

Non-Technical Clubs

- National Service Scheme (NSS): Member
- Bengali Club: Discipline Co-Lead

Creative Pursuits:

- Content Writing: Conducted in-depth research to create engaging and informative content for a variety of audiences.
- Journaling: Maintained detailed research logs and reflective writings.
- Dance: 18 years of experience in classical and semi-classical dance and won award under the famous PadmaShri award winner Smt. Mamata Shankar.

Competitions:

- Pathomath (Mood Indigo): Participated in this prestigious mathematical contest, showcasing analytical problem-solving abilities.

ACHIEVEMENTS & RECOGNITION

- **Research Symposium Participant:** Shared findings on biomechanical impact studies at university-level events.
- **Communication Recognition:** Acknowledged for effective communication during university and club activities.

PUBLIC HEALTH & EPIDEMIOLOGY ASPIRATIONS

- Keen on addressing mental health disparities and developing AI-powered healthcare solutions to combat public health crises. Passionate about conducting epidemiological studies to understand disease patterns and improve preventive healthcare measures.
- Passionate about developing and optimizing healthcare solutions using biomedical technologies and AI-driven analytics.

HOBBIES & INTERESTS

- **Reading Novels:** Enthusiast of contemporary fiction and scientific literature.
- **Creative Writing:** Occasional novelist with a passion for storytelling.
- **Community Engagement:** Actively involved in volunteering for public health awareness campaigns.

LANGUAGES:

English, Bengali, Hindi